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CAMBRIDGE
INTERNATIONAL EXAMINATIONS

June 2003

INTERNATIONAL GCSE

MARK SCHEME

MAXIMUM MARK: 80

SYLLABUS/COMPONENT: 0625/03

PHYSICS
Paper 3 (Extended)

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1	**(a)**	**(i)**	force of gravity acts on masses/weight of masses	**B1**	
		(ii)	vector has direction/force has direction	**B1**	**2**
	(b)	**(i)**	spring 1 (more difficult)	**M1**	
			any correct relevant pair of values	**A1**	
		(ii)	P marked at extension 25 mm to 28 mm	**A1**	
			explanation in terms of end of proportionality	**B1**	
		(iii)	each graph read at 15 N, approx. 25 mm, 19 mm	**C1**	
			difference correct, 6 mm +/- 1 mm	**A1**	**6**
					[8]
2	**(a)**		change in speed is 1.5 m/s	**C1**	
			deceleration = decrease in speed/time or 1.5/12	**C1**	
			a = (-/+) 0.125 m/s	**A1**	**3**
	(b)		average speed = 1.75 m/s	**C1**	
			distance = 21 m	**A1**	**2**
					[5]
3	**(a)**		attempt to use triangle or parallelogram of forces	**M1**	
			stated scale used	**A1**	
			950 N and 1220 N in correct relative directions	**C1**	
			correct resultant drawn in	**C1**	
			weight = 1785 N [limits 1700 N to 1850 N]	**A1**	**5**
	(b)	**(i)**	work = force x distance or 1500 x 3.0	**C1**	
			work = 4500 J	**A1**	
		(ii)	power = work/time or 4500/2.5	**C1**	
			power = 1800 W	**A1**	**4**
					[9]
4	**(a)**		air molecules hit dust particles	**M1**	
			hits continuously/unevenly/hits cause movement in all directions	**A1**	
			air molecules fast moving/high energy	**B1**	**3**
	(b)		any attempt to use p x v = constant or correct proportion	**C1**	
				C1	
			fraction 2 x 80/25 seen	**A1**	**3**
			p = 6.4 x 10 (Pa)		
					[6]
5	**(a)**		Y is a wire of different metal/not copper	**B1**	
			Z is a galvanometer/millivoltmeter/milliammeter	**B1**	**2**
	(b)		2 junctions at different temperatures, accept one hot, one cold	**B1**	
			temperature difference causes e.m.f./voltage/current	**B1**	
			reading of meter changes (with temperature)	**B1**	
			1 junction at known temperature/need for calibration	**B1**	**max 3**
	(c)		dull or black surface	**B1**	**1**
					[6]

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6	**(a)**	**(i)**	incident ray, refracted ray and normal drawn	**C1**	
			all correct and meeting at a point	**A1**	
		(ii)	angle of incidence and refraction correctly identified	**B1**	
		(iii)	values correct within agreed limits	**B1**	**4**
	(b)		use of sini/sinr	**C1**	
			correct substitution from candidates values	**C1**	
			value correct within agreed limits from candidate's values	**A1**	**3**
					[7]
7	**(a)**		value 3 x 10 m/s	**A1**	**1**
	(b)		speed of light (much) greater than speed of sound or value for sound	**A1**	**1**
	(c)	**(i)**	source and receiver arrangement	**C1**	
			with detail and labels	**A1**	
		(ii)	distance between source and receiver	**B1**	
			time between flash and bang	**B1**	
		(iii)	speed = distance/time	**B1**	**max 4**
					[6]
8	**(a)**	**(i)**	use of charge = It or I = 90/45	**C1**	
			current = 2 A	**A1**	
		(ii)	resistance = voltage/current or 6/2	**C1**	
			resistance is 3 ohm	**A1**	
		(iii)	energy = Vit or Vq or 6 x 90	**C1**	
			energy is 540 J	**A1**	**6**
	(b)		idea of energy transfer	**C1**	
			is (6) J/C	**A1**	**2**
					[8]
9	**(a)**	**(i)**	power = VI or 24 X 2	**C1**	
			power is 48 W	**A1**	
		(ii)	voltage = power/current or 48/0.4	**C1**	
			voltage is 120 V	**A1**	**4**
	(b)	**(i)**	no/very little energy/power lost or energy/power in = energy/power out	**B1**	
		(ii)	any mention of magnetic field	**B1**	
			changing magnetic field	**B1**	
			field passes through core or secondary coil	**B1**	
			induces voltage in secondary coil	**B1**	
			number of turns on secondary determines voltage output	**B1**	**max 4**
					[8]

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10	**(a)**	**(i)**	circular line of force around wire through P	**M1**	
			arrow(s) on line anticlockwise - none wrong	**A1**	
		(ii)	arrow through Q to left	**A1**	**3**
	(b)	**(i)**	none/stays same	**B1**	
		(ii)	direction reverses	**B1**	**2**
	(c)		at S - stronger	**B1**	
			at T - same (strength)	**B1**	
			at W - same (strength)	**B1**	**3**
					[8]
11	**(a)**	**(i)**	source, detector	**B1**	
			named absorber/air and labels	**B1**	
		(ii)	take detector reading with no source (background)	**B1**	
			detector reading with source, detector and air only	**B1**	
			detector reading with appropriate named absorber (including distance in air)	**B1**	
		(iii)	same reading with absorber(including air) as background	**B1**	
			so all alpha absorbed by cardboard/paper/air, others would get through	**B1**	**max 6**
	(b)		curved path stated or drawn	**B1**	
			path at right angles to magnetic field	**B1**	
			into paper	**B1**	**3**
					[9]

TOTAL 80